

MMS DATABASE ARCHITECTURE & SUPABASE SCHEMA

Enterprise Data Structure Specification

Version 1.0

PURPOSE

This document defines the complete data architecture for the MMS Digital Ecosystem.

The database must support:

Student Management

Applications

Courses

Learning Management

Certificates

Payments

Documents

Notifications

AI Systems

Reporting

Administration

Future Growth

DATABASE ENGINE

Supabase PostgreSQL

Requirements:

Row Level Security

Realtime Capabilities

Backups

Scalability

Audit Logging

USER ROLES

Student

Instructor

Administrator

Super Administrator

Future Employer

Future Partner Institution

TABLE: USERS

Purpose:

Master user account table.

Fields:

id

email

password_hash

role

first_name

last_name

phone

whatsapp

profile_image

status

created_at

updated_at

TABLE: STUDENTS

Purpose:

Store student-specific information.

Fields:

id

user_id

student_number

id_number

date_of_birth

gender

nationality

address

city

province

postal_code

employment_status

emergency_contact_name

emergency_contact_phone

status

created_at

updated_at

TABLE: APPLICATIONS

Purpose:

Track all applications.

Fields:

id

application_number

student_id

course_id

intake_id

application_status

submission_date

review_date

reviewed_by

comments

created_at

updated_at

APPLICATION STATUS TYPES

Draft

Submitted

Pending Review

Documents Required

Approved

Rejected

Cancelled

Completed

TABLE: COURSES

Purpose:

Course catalogue.

Fields:

id

course_name

course_code

description

price

duration

category

image_url

is_active

created_at

updated_at

TABLE: COURSE_MODULES

Purpose:

Learning content structure.

Fields:

id

course_id

module_title

module_description

module_order

created_at

TABLE: LESSONS

Purpose:

Learning content.

Fields:

id

module_id

lesson_title

lesson_type

content

video_url

resource_url

duration

lesson_order

created_at

LESSON TYPES

Video

PDF

Presentation

Quiz

Assignment

External Resource

TABLE: ENROLLMENTS

Purpose:

Student registrations.

Fields:

id

student_id
course_id
enrollment_date
start_date
end_date
access_expiry
enrollment_status
payment_status
created_at
updated_at

ENROLLMENT STATUS

Pending
Active
Suspended
Completed
Expired
Cancelled

TABLE: PAYMENTS

Purpose:
Payment records.
Fields:

id

student_id

course_id

amount

payment_method

reference_number

payment_date

payment_status

proof_of_payment

verified_by

verified_at

created_at

PAYMENT STATUS

Pending

Awaiting Verification

Verified

Rejected

Refunded

TABLE: DOCUMENTS

Purpose:

Document management.

Fields:

id

student_id

document_type

document_name

file_url

uploaded_at

verified

verified_by

DOCUMENT TYPES

ID Copy

Proof Of Address

Proof Of Payment

Passport

Certificate

Application Form

Other

TABLE: CERTIFICATES

Purpose:

Certificate records.

Fields:

id

certificate_number

student_id

course_id

issue_date

expiry_date

verification_code

certificate_url

status

created_at

TABLE: QUIZZES

Purpose:

Knowledge assessments.

Fields:

id

course_id

quiz_name

description

passing_score

created_at

TABLE: QUIZ_QUESTIONS

Fields:

id

quiz_id

question

question_type

correct_answer

points

created_at

TABLE: QUIZ_RESULTS

Fields:

id

student_id

quiz_id

score

passed

attempt_number

completed_at

TABLE: STUDENT_PROGRESS

Purpose:

Track learning activity.

Fields:

id

student_id

lesson_id

completed

completion_percentage

time_spent

completed_at

TABLE: ANNOUNCEMENTS

Purpose:

Broadcast communication.

Fields:

id

title

message

audience

created_by

created_at

TABLE: NOTIFICATIONS

Purpose:

Personal notifications.

Fields:

id

user_id

title

message

notification_type

is_read

created_at

TABLE: EMAIL_LOGS

Purpose:

Track all emails.

Fields:

id

recipient

subject

email_type

status

sent_at

provider

TABLE: WHATSAPP_QUEUE

Purpose:

Track follow-up requirements.

Fields:

id

student_id

assigned_to

message_type

status

scheduled_at

completed_at

STATUS TYPES

Pending

Assigned

Contacted

Completed

Cancelled

TABLE: SUPPORT_TICKETS

Purpose:

Student support.

Fields:

id

student_id

subject

message

status

assigned_to

created_at

updated_at

TABLE: AI_CONVERSATIONS

Purpose:

Store AI interactions.

Fields:

id

user_id

session_id

message

response

created_at

TABLE: INTAKES

Purpose:

Training schedules.

Fields:

id

intake_name

start_date

end_date

capacity

status

created_at

TABLE: INSTRUCTORS

Purpose:

Instructor profiles.

Fields:

id

user_id

employee_number

specialization

phone

email

status

created_at

TABLE: COURSE_ASSIGNMENTS

Purpose:

Link instructors to courses.

Fields:

id

course_id

instructor_id

assigned_date

status

TABLE: AUDIT_LOGS

Purpose:

Track all actions.

Fields:

id

user_id

action

table_name

record_id

timestamp

ip_address

AUTOMATIC SYSTEM EVENTS

Student Registers:

Create User

Create Student Record

Create Application

Generate Reference Number

Generate Temporary Login

Send Student Email

Send Instructor Email

Create WhatsApp Task

Create Notification

AUTOMATED EMAIL TRIGGERS

Application Submitted

Application Approved

Application Rejected

Payment Verified

Documents Requested

Course Starting

Course Completed

Certificate Issued

Password Reset

REPORTING MODULE

Total Students

Total Applications

Total Revenue

Course Performance

Enrollment Trends

Completion Rates

Instructor Performance

Marketing Performance

ROW LEVEL SECURITY

Students:

Access Own Data Only

Instructors:

Access Assigned Students

Administrators:

Access Operational Data

Super Administrators:

Full Access

FUTURE EXPANSION

Employer Portal

Recruitment System

Job Board

SMS Integration

WhatsApp Automation

Mobile Application

Multi-Campus Support

Accreditation Management

Equipment Management

HR Module

Finance Module

FINAL OBJECTIVE

The MMS database must function as the single source of truth for the entire institution.

Every application, payment, certificate, course, student, document, notification, and learning activity must be managed through this centralized system.

The architecture must support growth from a startup institution to a national training organization without requiring a complete redesign.